Task Management – Technical Documentation

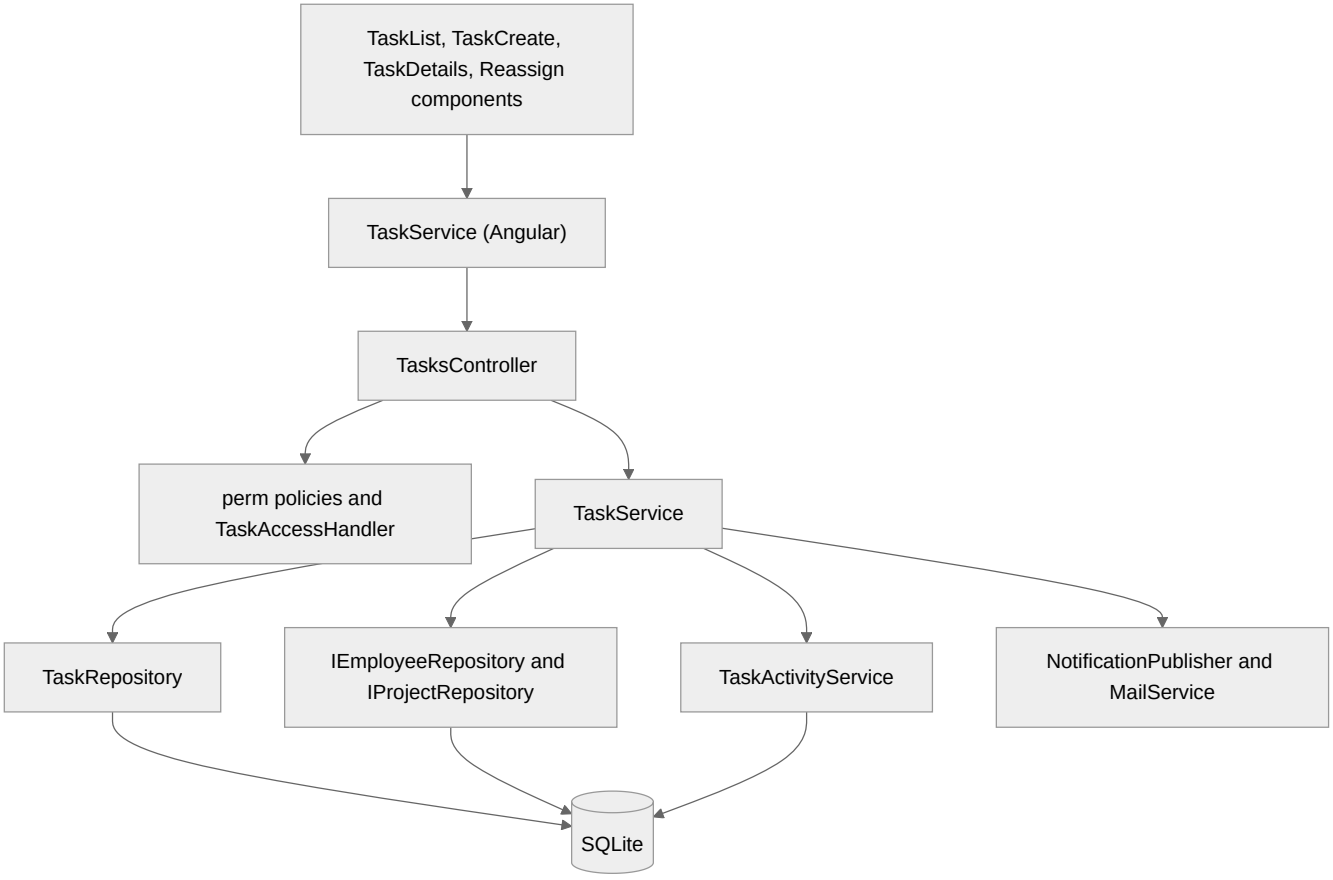
Project: TaskFlow – Intelligent Task Allocation & Tracking System
Parent Module: Monitoring · **Child Module:** 04.1 Task Management
Version: 1.0 · **Last Updated:** 31-Oct-2026
Backend: ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core)
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1. Technical Overview

Task Management is served by `TasksController` (`[Route("api/v1/tasks")]`, class-level `[Authorize]`), which exposes 12 endpoints. Three of them (`/board`, `/{id}/status`, `/assignees`) are also used by 04.3 Kanban Board.

- The controller reads the user id (`ClaimTypes.NameIdentifier`) and role name (`ClaimTypes.Role`) from the JWT and passes them to `TaskService`.
- `TaskService` enforces the task scope (Own, Team or Organization), the status-transition table, create / edit / reassign validation, worked-hours tracking, and notifications and internal mail.
- `TaskRepository` uses EF Core over SQLite. Tasks are soft-deleted (`IsDeleted`).
- Authorization: permission policies (`perm:<key>`), the `TaskAccess` resource policy on details, status and recipients, and the service-level team check for Manager-scoped roles.
- Frontend: standalone pages `TaskListComponent`, `TaskCreateComponent`, `TaskDetailsComponent` and `ReassignComponent`, calling the Angular `TaskService`.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Controller	`TaskFlow.Api/Controllers/Monitoring/`	`TasksController.cs`
Service	`TaskFlow.Api/Services/Monitoring/TaskManagement/Tasks/`	`ITaskService.cs`, `TaskService.cs`
Service rules	`TaskFlow.Api/Services/Monitoring/TaskManagement/Rules/`	`TaskAccessScope.cs`, `TaskStatusTransitions.cs`
Service (shared)	`TaskFlow.Api/Services/Monitoring/TaskManagement/Access/`	`ITaskAccessService.cs`, `TaskAccessService.cs`

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Layer	Path	Files
Repository	`TaskFlow.Api/Repositories/Monitoring/TaskManagement/Tasks/`	`ITaskRepository.cs`, `TaskRepository.cs`, `TaskListCriteria.cs`, `TaskVisibility.cs`, `TaskDateRange.cs`, `TaskBucketSelection.cs`
DTOs	`TaskFlow.Api/Dtos/Monitoring/TaskManagement/`	`TaskDtos.cs` (`TaskRecipientDto`, `TaskCardDto`, `TaskDetailsDto`, `TaskListItemDto`), `TaskRequests.cs` (`CreateTaskRequest`, `UpdateTaskRequest`, `UpdateTaskStatusRequest`, `ReassignTaskRequest`, `LogHoursRequest`, `SetRecipientsRequest`, `TaskListFilter`, `TaskBucketFilter`)
Validators	`TaskFlow.Api/Dtos/Monitoring/TaskManagement/Validators/`	`CreateTaskRequestValidator.cs`, `UpdateTaskRequestValidator.cs`, `UpdateTaskStatusRequestValidator.cs`, `ReassignTaskRequestValidator.cs`, `SetRecipientsRequestValidator.cs`
Entities	`TaskFlow.Api/Models/Entities/`	`Task.cs`, `TaskRecipient.cs`
Enums	`TaskFlow.Api/Models/Enums/`	`TaskStatus.cs`, `TaskPriority.cs`, `ReminderFrequency.cs`, `NotificationType.cs`
Shared rules	`TaskFlow.Api/Common/Rules/`	`TaskStageCalculator.cs`, `TaskBucketRules.cs`, `ReminderIntervals.cs`
Authorization	`TaskFlow.Api/Authorization/TaskAccess/`	`TaskAccessRequirement.cs`, `TaskAccessHandler.cs` (shared with 04.2 and 04.3)

4. Frontend File Structure

Type	Path	Purpose
Page component	`src/app/modules/monitoring/task-management/pages/`	`task-list.component.*` (Screen 7), `task-create.component.*` (Screen 8, including copy), `task-details.component.*` (Screen 9, with the 04.2 summary tabs for Screens 11/18), `reassign.component.*` (Screen 15)
Page model	`src/app/modules/monitoring/task-management/pages/task-create.model.ts`	Create-form field names (`FieldName`)
API service	`src/app/modules/monitoring/task-management/services/task.service.ts`	`list`, `getAssignees`, `create`, `getBoard`, `getById`, `update`, `logHours`, `changeStatus`, `delete`, `reassign`, `getRecipients`, `setRecipients` (the comment and activity calls belong to 04.2)
Models	`src/app/modules/monitoring/task-management/models/task.model.ts`	`TaskCard`, `TaskDetails`, `TaskListItem`, `PagedResult`, `TaskListFilter`, request types, `TASK_STATUSES`, `TASK_PRIORITIES`

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Type	Path	Purpose
Utils	`src/app/modules/monitoring/task-management/utils/return-url.ts`	`safeReturnUrl`, `backTarget` (Back to tasks / board / project)
Route / Guard	`src/app/modules/monitoring/monitoring.routes.ts` + `permissionGuard`	See the table below

Route	Guard permission
`/tasks`, `/tasks/:id`	none (signed-in only)
`/tasks/new`	`tasks.create`
`/tasks/:id/reassign`	`tasks.reassign`

5. API & Logic

5.1 API Endpoints

All routes are under `[Route("api/v1/tasks")]` with class-level `[Authorize]` (controller `TasksController`).

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
1	GET	`/api/v1/tasks`	Paged, filtered, searched list scoped to the caller	`[Authorize]` only (scope applied in service)	`TaskListFilter` (query)	`PagedResult<TaskListItemDto>`
2	POST	`/api/v1/tasks`	Create task	`perm:tasks.create` + service check	`CreateTaskRequest`	`201 Created` + `TaskDetailsDto` (Location → GetById)
3	GET	`/api/v1/tasks/assignees`	Distinct assignees of accessible tasks (list/board filter)	`[Authorize]` only	–	`IReadOnlyCollection<TaskRecipientDto>`
4	GET	`/api/v1/tasks/board`	Flat list of accessible tasks for the board	`perm:board.view`	–	`IReadOnlyCollection<TaskCardDto>`
5	GET	`/api/v1/tasks/{id:int}`	Task details	`TaskAccess` + service team check	–	`TaskDetailsDto` (404 when outside the manager's team)
6	PUT	`/api/v1/tasks/{id:int}`	Edit task	`perm:tasks.update` + service scope	`UpdateTaskRequest`	`TaskDetailsDto`

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#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
7	POST	`/api/v1/tasks/{id:int}/hours`	Add worked hours	`[Authorize]` only; service: assignee or `tasks.update` + scope	`LogHoursRequest`	`TaskDetailsDto`
8	DELETE	`/api/v1/tasks/{id:int}`	Soft-delete task	`perm:tasks.delete`	–	`204 No Content`
9	PATCH	`/api/v1/tasks/{id:int}/status`	Change status (transition table)	`perm:tasks.status.change` + `TaskAccess` + service scope	`UpdateTaskStatusRequest`	`TaskCardDto`
10	POST	`/api/v1/tasks/{id:int}/reassign`	Reassign with reason	`perm:tasks.reassign` + service check and scope	`ReassignTaskRequest`	`TaskDetailsDto`
11	GET	`/api/v1/tasks/{id:int}/recipients`	Task recipients	`TaskAccess` + service scope	–	`IReadOnlyCollection<TaskRecipientDto>`
12	PUT	`/api/v1/tasks/{id:int}/recipients`	Replace additional recipients	`perm:tasks.recipients.manage` + service check	`SetRecipientsRequest`	`204 No Content`

Rows 3, 4 and 9 are also the endpoints of 04.3 Kanban Board.

`TaskListFilter` query parameters:

- `assignedToUserId`, `status`, `priority`, `deadlineFrom`, `deadlineTo`, `search`, `page` (default 1), `pageSize` (default 20; values outside 1–100 become 20).
- Dashboard drill-down parameters: `bucket` (`All`, `Pending`, `InProgress`, `Completed`, `Overdue`), `deadlineBucket`, `departmentId`, `projectId`, `employeeId`, `managerId`, `from`, `to` (`DateOnly`), `basis` (`AnalyticsDateBasis`, default `Created`).
- Enums travel as their names (`JsonStringEnumConverter`).

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5.2 Request / Response Samples

```
// POST /api/v1/tasks
{
  "title": "Prepare Q4 release notes",
  "description": "Collect changes from all teams",
  "assignedToUserId": 12,
  "projectId": 3,
  "priority": "High",
  "startDate": "2026-10-31T04:30:00Z",
  "deadline": "2026-11-03T12:30:00Z",
  "estimatedHours": 6,
  "additionalRecipientUserIds": [7, 9],
  "firstReminderDays": 2,
  "secondReminderDays": 1,
  "overdueReminderFrequency": "Daily"
}
```

```
// 201 Created
{
  "id": 42, "taskCode": "TSK-000042", "title": "Prepare Q4 release notes",
  "description": "Collect changes from all teams",
  "assignedToUserId": 12, "assignedToName": "Asha Nair",
  "createdByUserId": 4, "createdByName": "Ravi Menon",
  "priority": "High", "startDate": "2026-10-31T04:30:00Z", "deadline": "2026-11-03T12:30:00Z",
  "estimatedHours": 6, "actualHours": null, "status": "ToDo", "stage": "Upcoming",
  "recipients": [ { "userId": 7, "fullName": "Divya K" }, { "userId": 9, "fullName": "John P"
}, { "userId": 12, "fullName": "Asha Nair" } ],
  "createdAt": "2026-10-31T10:00:00Z", "updatedAt": "2026-10-31T10:00:00Z",
  "projectId": 3, "projectName": "Website Revamp",
  "firstReminderDays": 2, "secondReminderDays": 1, "overdueReminderFrequency": "Daily"
}
```

```
// GET /api/v1/tasks?status=InProgress&search=release&page=1&pageSize=20  -> 200 OK
{ "items": [ { "id": 42, "taskCode": "TSK-000042", "title": "Prepare Q4 release notes",
"assignedToUserId": 12,
  "assignedToName": "Asha Nair", "createdByUserId": 4, "createdByName": "Ravi Menon",
"priority": "High",
  "startDate": "2026-10-31T04:30:00Z", "deadline": "2026-11-03T12:30:00Z", "estimatedHours": 6,
  "status": "InProgress", "stage": "Upcoming", "isOverdue": false, "projectId": 3,
"projectName": "Website Revamp" } ],
  "total": 1, "page": 1, "pageSize": 20 }
```

```
// PATCH /api/v1/tasks/42/status
{ "status": "InQA" }
// POST /api/v1/tasks/42/reassign
{ "newAssigneeUserId": 15, "reason": "Asha is on leave next week" }
// POST /api/v1/tasks/42/hours
{ "hours": 2.5 }
// PUT /api/v1/tasks/42/recipients
{ "recipientUserIds": [7, 9] }
```

> The `PagedResult` property names (`items`, `total`, `page`, `pageSize`) are taken from the frontend `PagedResult<T>` model and from the constructor `new PagedResult<TaskListItemDto>(items,`

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> `result.Total, page, pageSize)`. The sample values are illustrative.

5.3 Service Layer Logic

`TaskService` (`Services/Monitoring/TaskManagement/Tasks/TaskService.cs`).

Shared helpers

- `HasPermissionAsync(role, key)` reads the role's keys from the database (`IRoleRepository.GetPermissionKeysByRoleNameAsync`) and expands legacy keys (`EffectivePermissions.Expand`). Role changes therefore apply at once.
- `IsManagerScopedAsync(role)` = holds `dashboard.view.team` and does not hold `dashboard.view.organization`.
- `VisibilityForCallerAsync(userId, role)`:
  - No `tasks.view.all`, or not manager-scoped → `TaskVisibility(userId, canViewAll)`.
  - Manager-scoped with `tasks.view.all` → `TaskVisibility(userId, ViewAll: true, TeamUserIds: GetTeamMemberIdsAsync(userId), OrOwn: true)`.
  - The team is the active users, other than the manager, in a department the manager manages (`DepartmentManagers`) or on a project where the manager has `IsProjectManager`.
- `CanAccessTaskAsync(task, recipientIds, userId, role)`:
  1. `TaskAccessScope.CanAccess` (view-all, assignee, creator or recipient).
  2. For a manager-scoped view-all caller who is not personally on the task, also `ITaskRepository.IsTeamTaskAsync` (assigned to, or notifying, a team member).

`ListAsync(TaskListFilter filter, int currentUserId, string currentUserRole, ...)`

1. Clamp `page` (minimum 1). A `pageSize` outside 1–100 becomes 20.
2. Build the visibility. If the filter is a drill-down (`Bucket`, `DeadlineBucket`, `DepartmentId`, `ProjectId`, `EmployeeId`, `ManagerId`, `From` or `To` is set) and the caller is manager-scoped, set `OrOwn = false`, so only the team's tasks are listed.
3. `DrillDownCriteriaAsync` converts:
   - the bucket to a `TaskBucketSelection`
   - the deadline bucket to `DeadlineBucketRules.BoundsFor(now, timeZone)`
   - `From` / `To` to UTC local-midnight bounds (`ReminderCalculator.LocalMidnightUtc`); if `From > To` it throws `AnalyticsFilterException("The end date cannot be before the start date.")` (400).
4. `ITaskRepository.ListAsync(criteria, page, pageSize)`. Each row is mapped to `TaskListItemDto`. The stage comes from `TaskStageCalculator.Calculate(status, deadline, now, appTimeZone)`, and `IsOverdue = stage == Overdue`.

`GetBoardAsync` / `GetAssigneesAsync` both use `VisibilityForCallerAsync`. The board maps each task to `TaskCardDto`, with `IsOverdue = Deadline < UtcNow` and status not Completed or Cancelled (see 04.3).

`GetDetailsAsync(taskId, ...)`

1. Load with `GetWithPeopleForUpdateAsync` (includes people even when soft-deleted).
2. Load the recipients.
3. `CanAccessTaskAsync`. If it fails, return `null` (the controller returns 404).
4. `ToDetails`, which includes the stage.

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`CreateAsync(CreateTaskRequest request, int currentUserId, string currentUserRole, ...)`

1. `tasks.create` check → `ForbiddenException("Your role is not allowed to create tasks.")`.
2. `ValidateCreate` (title, assignee > 0, project > 0, deadline after start, estimated hours > 0). This repeats the FluentValidation rules.
3. The assignee must be live and active (`IEmployeeRepository.GetLiveAsync`) → `"The assigned employee does not exist or is not active."`.
4. The project must exist and be active (`IProjectRepository.ExistsActiveAsync`) → `"The selected project does not exist."`.
5. Recipients are de-duplicated, and every one must be live (`CountLiveAsync`) → `"One or more recipients do not exist."`.
6. Reminder fields that are null take the defaults from `IReminderSettingsService.GetAsync` (`DefaultFirstDays`, `DefaultSecondDays`, `DefaultOverdueFrequency`). The order is checked with `ReminderIntervals.AreOrdered`.
7. In one `IUnitOfWork.ExecuteInTransactionAsync`:
    1. Insert the task with a placeholder `TaskCode` `P-{guid}`, `Status = ToDo` and `CreatedByUserId = caller`, and save.
    2. Set `TaskCode = "TSK-{Id:D6}"`.
    3. Add one `TaskRecipient` per user in `{assignee} ∪ additional` (BR-5), and save.
    4. Append the activity entries `Created` ("Task created", new value = title) and `Assigned` ("Assigned to {name}", new value = assignee id).
    5. Publish the notification `TaskAssigned` "A task was assigned to you: {title}". This is inside the transaction, so if it fails nothing is left behind.
8. Reload the task and return `TaskDetailsDto`.

`UpdateAsync(taskId, UpdateTaskRequest request, ...)`

1. `tasks.update` check → `"Your role is not allowed to edit tasks."`.
2. `ValidateUpdate`.
3. Load the task; not found → `null` (404). `CanAccessTaskAsync` fails → `ForbiddenException("You do not have access to this task.")`.
4. Append history entries for the fields that changed:
    - `DeadlineChanged` "Deadline updated" (old/new as ISO `O`)
    - `PriorityChanged` "Priority updated"
    - `HoursLogged` "Worked hours logged: {X}h (total)", when `ActualHours` is given and differs
5. Overwrite Title (trimmed), Description, Priority, StartDate, Deadline, EstimatedHours and UpdatedAt, then save.
6. If the hours changed, run `NotifyHoursChangedAsync`.

`LogHoursAsync(taskId, LogHoursRequest request, ...)`

1. `Hours` must be > 0 and ≤ 24 → `BusinessRuleException("Hours must be greater than 0 and at most 24 per entry.")`.
2. Load the task and check `CanAccessTaskAsync`.
3. The caller must be the assignee or hold `tasks.update` → `"Only the assignee can log hours on this task."`.
4. `ActualHours = previous + hours`. Append `HoursLogged` (old/new totals, written with `InvariantCulture` `0.##`) and save.

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5. `ActualHours` is a concurrency token. A `ConcurrencyConflictException` becomes `ConflictException("The task's worked hours changed while you were saving. Please try again.")` (409).
6. `NotifyHoursChangedAsync`.

`NotifyHoursChangedAsync` (best effort; a failure is only logged)

- Targets: the recipients plus the assignee, de-duplicated.
- Message: `"{actor} added Xh of worked time (was Yh) on task {title}. Total worked now: …h of …h estimated (…h remaining | over by …h)."` The wording changes to "cleared the worked hours" or "updated the worked hours" for those cases.
- Publishes `NotificationType.System` to each target, then sends internal mail through `IMailService.SendAsync(actorId, SendMailRequest(emails, null, "Worked hours changed: {title}", message, null))`.

`ChangeStatusAsync(taskId, UpdateTaskStatusRequest request, ...)`

1. Load the task and check `CanAccessTaskAsync`.
2. `TaskStatusTransitions.Can(from, to)`, otherwise `BusinessRuleException("Cannot move a task from {from} to {to}.")`.
3. Set Status and UpdatedAt. Set `CompletedAt` on Completed and `CancelledAt` on Cancelled.
4. Append `StatusChanged` ("Status changed from X to Y"), plus `Completed` ("Task completed") on completion. Save.
5. Return `TaskCardDto`.

`ReassignAsync(taskId, ReassignTaskRequest request, ...)`

1. `tasks.reassign` check → `"Your role is not allowed to reassign tasks."`.
2. A blank reason → `"A reassignment reason is required."`.
3. Load the task and check the scope.
4. The new assignee must be live and active → `"The target employee does not exist or is not active."`.
5. Set `AssignedToUserId`. Append `Reassigned` "Reassigned from {old} to {new}. Reason: {reason}" with old/new names.
6. Add the new assignee as a recipient if missing (BR-5), and save.
7. `NotifyReassignedAsync` (best effort):
    - Targets: recipients ∪ new assignee ∪ old assignee ∪ actor.
    - The new assignee gets "A task was reassigned to you: {title}. Reason: {reason}". Everyone else gets `"{actor} reassigned task "{title}" ({code}) from A to B. Reason: …"`. The type is `NotificationType.Reassigned`.
    - Mail subject: "Task reassigned: {title}".
    - The previous assignee **stays** a recipient.

`GetRecipientsAsync` returns the recipients, or null when the task is not found or out of scope (404).

`SetRecipientsAsync(taskId, SetRecipientsRequest request, ...)`

1. `tasks.recipients.manage` check → `"Your role is not allowed to manage task recipients."`.
2. Task not found → `false` (404).

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3. Desired = request ids ∪ {assignee}. Add the missing ones and remove the extra ones (never the assignee). Save.
4. Append `Assigned` "Task recipients updated".

There is **no** task-scope check and **no** check that the users exist (see section 12).

`DeleteAsync(taskId, currentUserId, ...)`

1. Load the live task (`GetLiveForUpdateAsync`). Not found → `false` (404).
2. In a transaction: set `IsDeleted = true` and UpdatedAt, save, and append `Deleted` "Task deleted" (old value = title). Comments and history are kept.

There is no scope check (see section 12).

Rules classes

- `TaskAccessScope.CanAccess(userId, canViewAll, task, recipientIds)`.
- `TaskStatusTransitions.Can(from, to)`:
  - same → same is always allowed
  - ToDo → InProgress, Blocked or Cancelled
  - InProgress → InQA, Blocked, Completed or Cancelled
  - Blocked → InProgress or Cancelled
  - InQA → InProgress, Completed, Blocked or Cancelled
  - Completed and Cancelled are terminal
- `TaskStageCalculator.Calculate`: Completed or Cancelled → `Completed`; otherwise it compares the deadline's calendar day with today in the application time zone and returns `Overdue`, `DueToday` or `Upcoming`.

5.4 Repository Layer

`TaskRepository : BaseRepository<Task>, ITaskRepository`

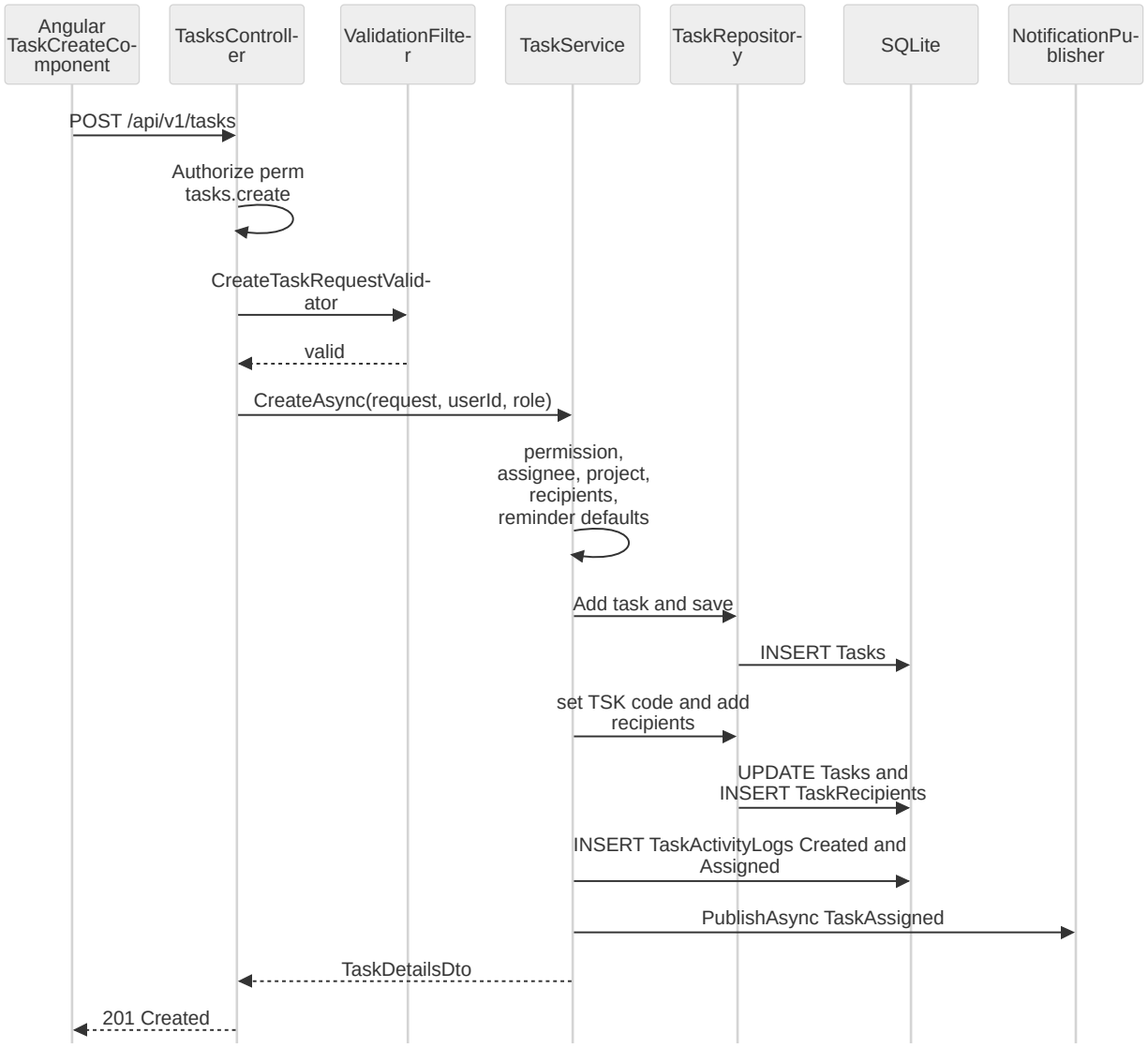
Method	Description	Query Notes
`ListAsync(TaskListCriteria, page, pageSize)`	One page of tasks	`Visible(visibility)`, then the filters: assignee, status, priority, deadline from/to, `LIKE` search on Title or TaskCode (with `%` `_` `\` escaped), bucket (`TaskBucketRules`), deadline bucket, date range (`ApplyDateRange`), entity filters (`ApplyEntityFilters`: department, project, employee, manager). `CountAsync`, then `IgnoreQueryFilters` + Include AssignedTo, CreatedBy, Project; ordered by Deadline then Id; Skip/Take
`GetBoardAsync(TaskVisibility)`	Every visible task	Include AssignedTo, Project; ordered by Deadline
`GetAssigneesAsync(TaskVisibility)`	Distinct assignees	Projection plus `Distinct`, ordered by name in memory

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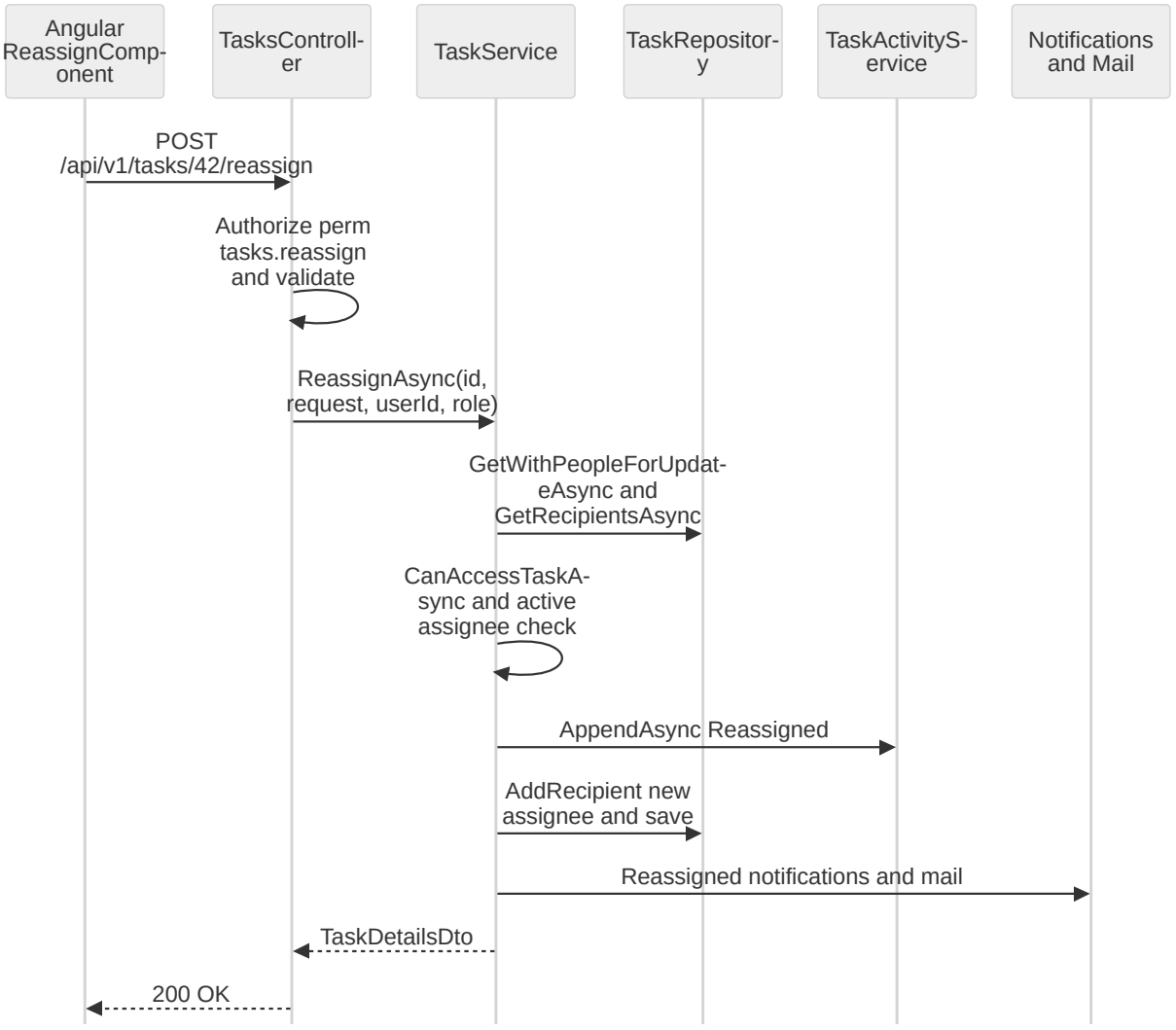
Method	Description	Query Notes
`IsTeamTaskAsync(taskId, teamUserIds)`	Is the task assigned to, or notifying, a team member	`TeamTasks` + `AnyAsync`
`GetWithPeopleForUpdateAsync(taskId)`	Tracked live task with people	`IgnoreQueryFilters` so soft-deleted users still show
`GetLiveForUpdateAsync(taskId)`	Tracked live task	`!IsDeleted`
`GetLiveAsync(taskId)`	Read-only live task	`AsNoTracking`
`GetRecipientsAsync(taskId)`	Recipients with names	`AsNoTracking`, ordered by UserId
`GetRecipientRowsForUpdateAsync(taskId)`	Tracked recipient rows	For replacing them
`GetRecipientIdsAsync(taskId)`	Recipient user ids	`AsNoTracking`
`AddRecipient` / `RemoveRecipients`	Stage inserts and deletes	–
`CountUnfinishedAssignedToAsync(userId)`	Live tasks not Completed or Cancelled	Used outside this module (employee management)
`Visible(TaskVisibility)` (private)	`Scoped(userId, viewAll)`; with `TeamUserIds`, limit to team tasks, or team plus own when `OrOwn`	`Scoped` = `!IsDeleted` and, without view-all, assignee, creator or recipient

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5.5 Sequence Diagrams



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6. Database Design

SQLite storage types come from the migrations: int and bool are stored as INTEGER, and string, DateTime and decimal as TEXT.

Table	Column	Type	Constraints	Description
`Tasks`	`Id`	int	PK, Identity	Task id
`Tasks`	`TaskCode`	string(32)	Not Null, Unique index	`TSK-000042` (placeholder `P-…` during the insert)
`Tasks`	`Title`	string(200)	Not Null	Title
`Tasks`	`Description`	string(4000)	Nullable	Description
`Tasks`	`AssignedToUserId`	int	FK → Users.Id, Not Null, Index	Assignee
`Tasks`	`CreatedByUserId`	int	FK → Users.Id, Not Null	Creator

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Table	Column	Type	Constraints	Description
`Tasks`	`ProjectId`	int	FK → Projects.Id, Nullable column (the API requires it on create)	Project
`Tasks`	`Priority`	enum `TaskPriority` (int)	Not Null	Low=0, Medium=1, High=2, Critical=3
`Tasks`	`StartDate`	datetime (UTC)	Not Null	Start
`Tasks`	`Deadline`	datetime (UTC)	Not Null, Index	Deadline
`Tasks`	`EstimatedHours`	decimal	Not Null	Estimate (> 0)
`Tasks`	`ActualHours`	decimal	Nullable, **concurrency token**	Worked total
`Tasks`	`Status`	enum `TaskStatus` (int)	Not Null	ToDo=0, InProgress=1, Blocked=2, Completed=3, Cancelled=4, InQA=5
`Tasks`	`FirstReminderDays`	int	Default 2 (entity)	First reminder interval
`Tasks`	`SecondReminderDays`	int	Default 1 (entity)	Second reminder interval
`Tasks`	`OverdueReminderFrequency`	enum `ReminderFrequency` (int)	Default Daily (entity)	Daily / Weekly
`Tasks`	`CreatedAt` / `UpdatedAt`	datetime	Not Null	Audit times
`Tasks`	`CompletedAt` / `CancelledAt`	datetime	Nullable	Set on those transitions
`Tasks`	`IsDeleted`	bool	Not Null	Soft delete
`TaskRecipients`	`Id`	int	PK	–
`TaskRecipients`	`TaskId`	int	FK → Tasks.Id; unique (`TaskId`, `UserId`)	Task
`TaskRecipients`	`UserId`	int	FK → Users.Id	Notified user (the assignee is always included)

Tables read but not owned: `Users` (assignee, creator, recipients), `Projects` (task project), `DepartmentManagers` and `UserProjectAllocations.IsProjectManager` (a Manager's team, through `IEmployeeRepository.GetTeamMemberIdsAsync`), `Roles` / `RolePermissions` (permission checks). Writes `TaskActivityLogs` (owned by 04.2).

Relationships

- `Tasks.AssignedToUserId → Users.Id` (FK, Restrict)
- `Tasks.CreatedByUserId → Users.Id` (FK, Restrict)

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- `Tasks.ProjectId → Projects.Id` (FK, Restrict)
- `TaskRecipients.TaskId → Tasks.Id` (FK, Cascade); `TaskRecipients.UserId → Users.Id` (FK, Cascade)

7. Security & Authorization

- **Authorization:** `perm:<key>` policies on create, edit, delete, status, reassign and set-recipients; the `TaskAccess` policy on details, status and get-recipients; `TaskService` re-checks the key and applies the Manager team scope.
- **Data scope:** Organization / Team / Own as resolved by `TaskService.VisibilityForCallerAsync` and `CanAccessTaskAsync` (see the parent document, section 7).
- **Frontend:** `permissionGuard` on `/tasks/new` and `/tasks/:id/reassign`; buttons gated by `canCreate`, `canEdit`, `canDelete`, `canReassign`.

Permission Key	Endpoint(s)	UI Element Controlled
`tasks.view.own` (Always On)	`GET /tasks`, `/tasks/board`, `/tasks/{id}`, `/tasks/assignees` (data scope)	Task List "My Tasks"
`tasks.view.all`	Scope in `TaskService`; `TaskAccessHandler` claim	"TaskFlow Monitoring" menu, "Task Management" title, Employee filter (list), Assignee filter (board), `getAssignees()` call
`tasks.create`	`POST /tasks`	Create task button, empty-state CTA, Copy (list row and details), `/tasks/new` guard
`tasks.update`	`PUT /tasks/{id}`; allows `POST /tasks/{id}/hours` for non-assignees	Edit task button, Time tracking panel (with the assignee)
`tasks.delete`	`DELETE /tasks/{id}`	Delete (list row and details)
`tasks.status.change`	`PATCH /tasks/{id}/status`	Board drag-and-drop (`canMove`). The details "Move to" buttons are **not** gated
`tasks.reassign`	`POST /tasks/{id}/reassign`	Reassign task tab, `/tasks/:id/reassign` guard
`tasks.recipients.manage`	`PUT /tasks/{id}/recipients`	none (no UI caller)

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
Title empty / over 200 characters (create, update)	`CreateTaskRequestValidator` / `UpdateTaskRequestValidator`	400 (validation problem)	FluentValidation default for `Title`
Description over 4000 characters	Validators	400	FluentValidation default
Assignee missing	`CreateTaskRequestValidator` / `TaskService`	400	An assigned employee is required.

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Rule	Layer	HTTP Code	Error Message
Project missing	`CreateTaskRequestValidator` / `TaskService`	400	A project is required.
Deadline not after start	Validators / `TaskService`	400	Deadline must be after the start date.
Estimated hours ≤ 0	Validators / `TaskService`	400	Estimated hours must be a positive number.
More than 50 additional recipients	`CreateTaskRequestValidator`	400	A task can have at most 50 additional recipients.
Reminder days outside 0–30 / wrong order	`CreateTaskRequestValidator` / `TaskService`	400	The second reminder cannot come before the first reminder.
Priority / status / frequency not in enum	Validators	400	FluentValidation default
Assignee inactive or missing	`TaskService.CreateAsync`	400	The assigned employee does not exist or is not active.
Project missing	`TaskService.CreateAsync`	400	The selected project does not exist.
Recipient missing	`TaskService.CreateAsync`	400	One or more recipients do not exist.
Status transition not allowed	`TaskService.ChangeStatusAsync`	400	Cannot move a task from {from} to {to}.
Reason missing / over 500 characters	`ReassignTaskRequestValidator` / `TaskService`	400	A reassignment reason is required.
New assignee inactive	`TaskService.ReassignAsync`	400	The target employee does not exist or is not active.
Hours outside 0–24 (log hours)	`TaskService.LogHoursAsync`	400	Hours must be greater than 0 and at most 24 per entry.
Recipients list null, more than 50, or id ≤ 0	`SetRecipientsRequestValidator`	400	FluentValidation default
Dashboard drill-down dates reversed	`TaskService.DrillDownCriteriaAsync`	400	The end date cannot be before the start date.
Role lacks the key (service re-check)	`TaskService`	403	Your role is not allowed to create tasks. / … edit tasks. / … reassign tasks. / … manage task recipients.
Out of scope (mutations)	`TaskService`	403	You do not have access to this task.
Not the assignee and no `tasks.update`	`TaskService.LogHoursAsync`	403	Only the assignee can log hours on this task.
Concurrent worked-hours change	`TaskService.LogHoursAsync`	409	The task's worked hours changed while you were saving. Please try again.

Shared rows (policy 403, 404 not found, generic 409 and 500) and the error format are in the parent document, section 8.

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9. Integrations / Side Effects

- **Notifications** (`INotificationPublisher.PublishAsync`; skipped when in-app notifications are switched off in Application Settings):
  - Create → `TaskAssigned` to the assignee (inside the create transaction).
  - Reassign → `Reassigned` to the new assignee, previous assignee, recipients and actor.
  - Worked hours changed (log hours or edit) → `System` to the assignee and recipients.
- **Email / internal mail:** reassign and hours changes use `IMailService.SendAsync(actorId, SendMailRequest(...))`, with subjects "Task reassigned: {title}" and "Worked hours changed: {title}". Best effort (logged, never rolled back).
- **Activity Log** (`TaskActivityService.AppendAsync`, owned by 04.2):

Action	Event type(s)
Create	`Created`, `Assigned`
Edit	`DeadlineChanged`, `PriorityChanged`, `HoursLogged`
Log hours	`HoursLogged`
Change status	`StatusChanged`, `Completed`
Reassign	`Reassigned`
Set recipients	`Assigned` ("Task recipients updated")
Delete	`Deleted`

- **Deadline reminders:** each task stores `FirstReminderDays`, `SecondReminderDays` and `OverdueReminderFrequency`. The reminder engine (Communication module) uses them.

> ⚠️ To be confirmed: that reminders stop for Completed or Cancelled tasks (BRD Screen 9 rule 9). This is implemented outside this module.

- **AI Assistant:**
  - `AssistantReadTools` (`get_my_tasks`, `prioritise_tasks`) call `ITaskService.ListAsync` with the caller's id and role, so they respect the task scope.
  - `AssistantManagerResolver` (used by `draft_mail`) calls `ITaskService.GetDetailsAsync` to find a task's creator.
- **Dashboard:** the drill-down filters reuse `TaskBucketRules`, `DeadlineBucketRules`, `ApplyDateRange` and `ApplyEntityFilters` from `Repositories/Dashboard/Analytics`, so a list opened from a chart holds exactly the tasks the chart counted.

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10. Configuration

Key (`appsettings.json`)	Purpose	Default
`Application:Timezone`	Fallback application time zone. Stage (Due today / Overdue), drill-down date bounds and deadline buckets use `IAppClock.GetTimeZoneAsync`	`Asia/Kolkata`
`Reminders:SchedulerEnabled` / `Reminders:TickSeconds`	Reminder engine that uses the task reminder fields	`true` / `60`

Settings stored in the database that this child reads:

- Reminder defaults (`IReminderSettingsService`)
- Default priority (frontend reads Application Settings)

> ⚠️ To be confirmed: whether `IAppClock` prefers the stored application-settings time zone over `Application:Timezone`.

The connection string, JWT lifetimes and the in-app notification switch are module-wide (parent document, section 10).

11. Testing

Paths are relative to `backend/tests`.

Test Type	File / Location	Covers
Integration	`TaskFlow.IntegrationTests/Monitoring/TaskManagement/TasksEndpointsTests.cs`	Task endpoints and authorization
Unit	`TaskFlow.UnitTests/Monitoring/TaskManagement/Tasks/TaskServiceTests.cs`, `TaskManagementServiceTests.cs`	Task service rules
Unit	`TaskFlow.UnitTests/Monitoring/TaskManagement/Tasks/CreateTaskRequestValidatorTests.cs`, `TaskValidatorsTests.cs`	Validators
Unit	`TaskFlow.UnitTests/Monitoring/TaskManagement/Tasks/CustomRolePermissionTests.cs`	Custom role permissions
Unit	`TaskFlow.UnitTests/Monitoring/TaskManagement/Tasks/ProjectTaskTests.cs`	Project on tasks
Unit	`TaskFlow.UnitTests/Monitoring/TaskManagement/Tasks/TaskDeletionTests.cs`	Soft delete
Unit	`TaskFlow.UnitTests/Monitoring/TaskManagement/Tasks/TaskLogHoursTests.cs`, `TaskWorkedHoursNotificationTests.cs`	Worked hours and notifications
Unit	`TaskFlow.UnitTests/Monitoring/TaskManagement/Rules/TaskTrackingDomainTests.cs`	Transitions, access scope, stage
Unit	`TaskFlow.UnitTests/Common/TaskBucketRulesTests.cs`	Bucket rules used by drill-down

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Test Type	File / Location	Covers
Support	`TaskFlow.UnitTests/TestSupport/TaskTrackingServiceHarness.cs`	Test harness

Only the test file names are available. What each file covers is inferred from its name.

12. Known Limitations / Gaps found

- The list's `projectId` filter is treated as a dashboard drill-down (`IsDrillDown`). For a Manager, picking a project in the normal list filter therefore removes the manager's own non-team tasks from the result (`OrOwn = false`).
- `task-list.component.ts` applies `Object.assign(request, this.drill()?.request)` twice. This is harmless duplicate code.
- The edit form sends `actualHours: 0` when the field is cleared, which records a worked-hours change. There is no validator rule against negative `ActualHours` in `UpdateTaskRequestValidator`.
- `UpdateTaskRequest` has no reminder fields, so reminders cannot be edited after creation.
- **`DELETE /tasks/{id}` has no task-scope check.** Any `tasks.delete` holder can delete any live task id (also listed in `permission-gaps.md` B4).
- **`PUT /tasks/{id}/recipients` has no task-scope check and does not check that the user ids exist.** `SetRecipientsAsync` checks only the permission and that the task exists. The audit (B6) says SetRecipients "relies on the service check", but none exists. There is no UI caller (B8).

  > ⚠️ To be confirmed: the HTTP result when a recipient id does not exist (FK failure on SQLite).

- **The Task Details "Move to" buttons are not gated by `tasks.status.change`** on the frontend (audit C). The API refuses the call with 403.
- **The Task List "Board view" button is not gated by `board.view`.** The route guard sends users without it back to the dashboard.
- **Hidden dependencies (audit D):**
  - Create Task needs `employees.view` (`UserService.getAll`), `projects.view` and Reminder Settings read access.
  - Reassign needs `employees.view`.
- **The Create Task assignee list is limited to `user.role === 'Employee'`** in `task-create.component.ts`, so custom-role users and Managers can never be picked in the UI (the API accepts any active user). The audit's same finding for `reassign.component.ts` is **stale**: reassign now lists any active user except the current assignee.
- **BRD vs code:** Screen 8 lists "Initial status". The code always creates tasks as To Do, with no field for it.
- **`tasks.update` catalog description** mentions "reminders", but the edit API and form do not change reminders (audit D, still valid).
- **Unreachable service messages:** the duplicated `ValidateCreate` / `ValidateUpdate` checks are normally pre-empted by FluentValidation (`ValidationFilter`), so users see the validator's message instead.

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